

Two-Sample z-Test for Means

Complete Line-by-Line Calculation Using Known Variances

No.	Group A	Group B
1	102	112
2	105	115
3	98	110
4	110	118
5	107	114
6	101	116
7	104	111
8	109	120
9	103	113
10	106	117
11	100	109
12	108	119

1. Given Values

Group A Mean = **104.4166667**

Group B Mean = **114.5**

Known Variance A = **14.0833**

Known Variance B = **13**

Observations: $n_A = 12$, $n_B = 12$

Hypothesized Mean Difference = **0**

Significance level $\alpha = 0.05$

2. Difference Between Means

Formula: $\bar{X}_A - \bar{X}_B$

= $104.4166667 - 114.5$

= **-10.0833333**

3. Standard Error

Formula: $SE = \sqrt{\sigma_A^2/n_A + \sigma_B^2/n_B}$

$SE = \sqrt{14.0833/12 + 13/12}$

$14.0833/12 = 1.173608$

$13/12 = 1.083333$

$SE = \sqrt{1.173608 + 1.083333}$

$SE = \sqrt{2.256941}$

$SE \approx 1.50231$

4. z-Statistic

Formula: $z = [(X_{\text{A}} - X_{\text{B}}) - \mu_{\text{A}}] / SE$

$\mu_{\text{A}} = 0$

$z = [(104.4166667 - 114.5) - 0] / 1.50231$

$z = -10.0833333 / 1.50231$

$z = -6.71188$

Excel value: **-6.711876505**

5. Why z is Negative

Group A mean is smaller than Group B mean:

$104.4167 - 114.5 = -10.0833$

Therefore z is negative.

For a two-tailed test, use $|z| = 6.71188$.

6. One-Tailed p-Value

Excel gives:

9.60687E-12

This is approximately 0.0000000000960687.

7. One-Tailed Critical z

For $\alpha = 0.05$:

z Critical one-tail = 1.644853627

8. Two-Tailed p-Value

Excel gives:

1.92137E-11

This is approximately 0.000000000192137.

9. Two-Tailed Critical z

For $\alpha = 0.05$, split α into two tails:

$\alpha/2 = 0.05/2 = 0.025$

z Critical two-tail = 1.959963985

10. Decision Using z

For the two-tailed test:

$|z| = 6.71188$

Critical z = 1.95996

$6.71188 > 1.95996$

Reject H_0 .

11. Decision Using p-Value

Two-tailed p-value = 1.92137E-11

$\alpha = 0.05$

$1.92137E-11 < 0.05$

Reject H_0 .

12. Hypothesis

Null hypothesis: $H_0: \mu_1 - \mu_2 = 0$

Alternative hypothesis: $H_1: \mu_1 - \mu_2 \neq 0$

13. Final Conclusion

There is a statistically significant difference between Group A and Group B.

Group A mean = 104.4167

Group B mean = 114.5

Difference = 10.0833 points

Group B has a higher average than Group A, and the difference is statistically significant at the 5% level.

Quick Formula Summary

Step	Formula / Item	Result
1	Mean A	104.4167
2	Mean B	114.5
3	Known Variance A	14.0833
4	Known Variance B	13
5	Standard Error	≈ 1.50231
6	z-statistic	-6.71188
7	One-tail p-value	9.60687E-12
8	Two-tail p-value	1.92137E-11
9	One-tail critical z	1.644853627
10	Two-tail critical z	1.959963985
11	Decision	Reject H_0